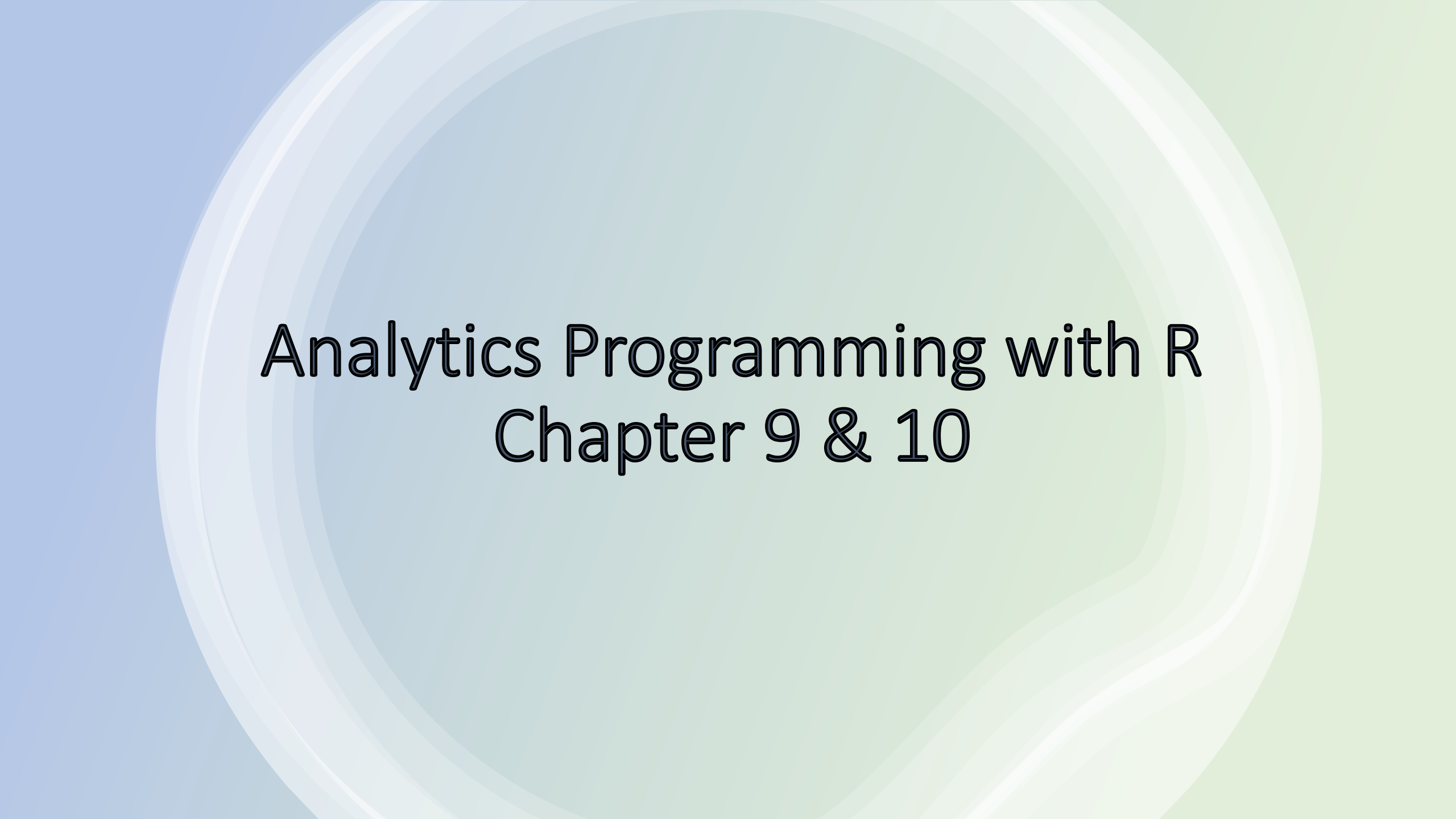




# Analytics Programming with R

## Chapter 9 & 10

# Control Statements

- if
- else
- ifelse
- switch

## An if statement with an else clause

```
if (score < 60) {  
  Grade <- "F"  
}
```

## An if statement with an else clause

```
if (score > 70) {  
  Grade <- "Satisfactory"  
} else {  
  Grade <- "Unsatisfactory"  
}
```

## An if statement with no braces that performs the same task

```
if (score > 70) Grade <- " Satisfactory " else Grade <- "  
Unsatisfactory "
```

## An if statement with two else if clauses and an else clause

```
if (score >= 90) {  
  Grade <- "A"  
} else if (score >= 80) {  
  Grade <- "B"  
} else if (score >= 70) {  
  Grade <- "C"  
} else {  
  Grade <- "F"  
}
```

if statements are similar to functions, all statements go inside the curly braces.

## A function for returning data from a function

```
return(value)
```

## A function that defines a parameter and returns a value

```
get_grade <- function(score) {  
  if (score > 70) {  
    Grade <- "Satisfactory"  
  } else {  
    Grade <- "Unsatisfactory"  
  }  
  return(Grade)  
}  
get_grade(80)  
[1] " Satisfactory "
```

## A nested if statement

```
if (Result == "Pass") {  
  if (Grade == "A") {  
    GPA <- 4  
  } else if (Grade == "B") {  
    GPA <- 3  
  }  
} else {  
  GPA <- 0  
}
```

## Another way to get the same result

```
if (Result == "Pass" & Grade == "A") {  
  GPA <- 4  
} else if (Result == "Pass" &  
           Grade == "B") {  
  GPA <- 3  
} else {  
  GPA <- 0  
}
```

## A function that defines two parameters and returns a value

```
get_GPA <- function(Result, Grade) {  
  if (Result == "Pass") {  
    if (Grade == "A") {  
      GPA <- 4  
    } else if (Grade == "B") {  
      GPA <- 3  
    }  
  } else {  
    GPA <- 0  
  }  
  return(GPA)  
}
```

```
get_GPA("Pass", "B")  
[1] 3
```

# Example of the Switch Statement

```
use.switch <- function(x)
{  switch(x,      "a"="first",      "b"="second",      "c"="last",
"d"="third",      "other")
}

# this is the default argument

use.switch("a")
use.switch("b")
use.switch("c")
use.switch("d")
use.switch("e")
```

## Example of the ifelse Statement

```
newNum <- c(4,4,3,2,NA)
ifelse(newNum == 4,"Yes","NO")
newNum <- c(4,4,3,2,NA)
ifelse(newNum == 4,"Yes","NO")
```

# Example of for Loop Statement

```
myCars <- c("Mustang", "Corvette", "Bently", "Porsche", "Maybach", "Ferrari")
```

```
# make a variable to hold their lengths with NA's using the rep() function to  
# replicate the NA values in myCars into myCarsLength to show NA in each element
```

```
myCarsLength <- rep(NA, length(myCars))  
myCarsLength
```

```
## [1] NA NA NA NA NA NA
```

```
# Use the names() function to assign the car names to the elements in myCarsLength
```

```
names(myCarsLength) <- myCars  
names(myCarsLength) <- myCars
```

```
# Loop through assigning the length of the characters of the car names to each element  
for (i in myCars)  
{  
  myCarsLength[i] <- nchar(i)  
}
```

# Example of for While Loop Statement

```
x <- 21
while(x <= 49)
{
  print(x)
  x <- x + 1
}
```

*# Using the next statement to skip over an iteration*

```
x <- 21
while(x <= 49)
{
  x <- x + 1
  print(x)

  if (x == 32)
  {
    next() # Skips over number 32 in the iteration
  }
}
```

*# Using the break statement to break out of a loop*

```
for(i in 1:10)
{
  if(i == 4)
  {
    break
  }
  print(i)
}
```